

ALGO EXAM CHEAT SHEET

1. Time Complexity

- Common Big-O: 1, $\log n$, n , $n \log n$, n^2 , 2^n , $n!$
- Rules: Drop constants, highest degree dominates

2. Sorting Algorithms

Insertion Sort: Best $O(n)$, Worst $O(n^2)$

Merge Sort: $O(n \log n)$, stable

Quick Sort: Avg $O(n \log n)$, Worst $O(n^2)$

Heap Sort: $O(n \log n)$

Counting Sort: $O(n + k)$

Radix Sort: $O(d(n + k))$

Bucket Sort: Avg $O(n)$

3. Recursion & Master Method

$$T(n) = aT(n/b) + f(n)$$

Case 1: $f(n)$ smaller $\rightarrow \Theta(n^{\log_b a})$

Case 2: equal $\rightarrow \Theta(n^{\log_b a} \log n)$

Case 3: larger $\rightarrow \Theta(f(n))$

4. String Matching

Naive: $O(nm)$

KMP: $O(n + m)$

Rabin-Karp: Avg $O(n + m)$, Worst $O(nm)$

5. Graph Algorithms

BFS: $O(V+E)$

DFS: $O(V+E)$

Dijkstra: $O((V+E) \log V)$

Bellman-Ford: $O(VE)$

Floyd-Warshall: $O(V^3)$

MST (Prim/Kruskal): $O(E \log V)$

6. Dynamic Programming

Rod Cutting: $O(n^2)$

Matrix Chain: $O(n^3)$

LCS: $O(nm)$

0/1 Knapsack: $O(nW)$

7. Greedy

Activity Selection: $O(n \log n)$

Huffman Coding: $O(n \log n)$

Fractional Knapsack: $O(n \log n)$

0/1 Knapsack $\rightarrow$ greedy fails